

Knicks Win Chaos

Common Knowledge, Coordination, and Equilibrium Routing
after a Championship at Madison Square Garden

Challenge 2 Submission

Significance. When the Knicks win, tens of thousands of fans pour out of Madison Square Garden into streets that close unpredictably. Getting everyone home is a problem of *information* and of *coordination*. Following De Freitas, Thomas, DeScioli, and Pinker [1], we show that what matters (as Schelling [5] anticipated) is not only what each person knows, but what they know that others know. We prove that (i) official alerts, social media, and public reports fuse into one closure map by adding log-odds; (ii) selfish routing on a *shared* map has a unique equilibrium that costs at most $4/3$ of a perfectly coordinated evacuation; (iii) a coordinated plan (a designated corridor plus extra trains) works in equilibrium *if and only if* the plan is approximately common knowledge, which only public broadcasts create; and (iv) crowd reporting of closures is a volunteer’s dilemma whose outcome zigzags with the level of knowledge, so bigger crowds report less unless the app breaks the symmetry. A Python prototype implements the model with a crowd-density map and halves pedestrian exposure to crush-level densities compared with naive routing.

Abstract

We model post-game egress as a Bayesian congestion game on a street graph, with closures as a random state and three information channels (official, social, crowd) as signals. We prove existence, uniqueness, and a $4/3$ price-of-anarchy bound for equilibrium routing on a public posterior map, and an exponential bound on the cost of misinformation. We then embed the stag-hunt and volunteer’s-dilemma analyses of [1] in this setting: coordinated egress plans are sustainable exactly on events that are common p^* -belief, and closure reporting follows the private/secondary/tertiary knowledge zigzag. A density-aware implementation is evaluated on a synthetic Midtown network.

Keywords: common knowledge | coordination games | Wardrop equilibrium | volunteer’s dilemma | crowd routing

1 Introduction

De Freitas et al. [1] argue that successful coordination requires *common knowledge*: everyone knows X , everyone knows that everyone knows X , and so on. They distinguish *private* knowledge (I know), *secondary* knowledge (I know you know), *tertiary* knowledge (I know you know I know), and common knowledge, typically created by a public broadcast such as a loudspeaker. In a butcher–baker stag hunt, participants cooperated more as knowledge rose, and most with common knowledge. In a bystander game, helping zigzagged across knowledge levels as predicted by the volunteer’s dilemma.

A championship night is the same structure at city scale. Fans must choose routes (a congestion game), the city and transit agencies must decide where to open corridors and add service (a coordination game), and someone has to report each closure (a volunteer’s dilemma). The challenge

asks whether official alerts, social media, and public reports can map closures and plan routes home. We answer yes, and identify the knowledge conditions under which each part works.

2 Model

Network and demand. Let $G = (V, E)$ be a directed street graph around the arena, K the set of origin–destination pairs (arena exits to transit hubs), $d_k > 0$ the mass of travelers for pair k , $D = \sum_k d_k$, and $\mathcal{P}_k$ the simple paths for pair k . A flow $f = (f_P)$ has $f_P \geq 0$ and $\sum_{P \in \mathcal{P}_k} f_P = d_k$; edge loads are $x_e = \sum_{P \ni e} f_P$. The feasible set $\mathcal{F}$ is compact and convex.

Closures and costs. The state $\theta \in \{0, 1\}^E$ has $\theta_e = 1$ if street e is closed, with independent priors $\mathbb{P}(\theta_e = 1) = \pi_e \in (0, 1)$. Edge costs are

$$c_e(x_e; \theta) = a_e x_e + b_e + M\theta_e, \quad a_e > 0, \quad b_e \geq 0, \quad (1)$$

where M is the time lost reaching a barricade and detouring. Social cost is $C(f; \theta) = \sum_e x_e c_e(x_e; \theta)$.

Information channels. For each edge, three signals are conditionally independent given θ_e :

1. **Official alert** $O_e \in \{1, \emptyset\}$: issued with probability α if $\theta_e = 1$, never if $\theta_e = 0$.
2. **Social media** (n_e, k_e) : k_e of n_e posts claim closure; each does so with probability p if closed, q if open, $0 < q < p < 1$.
3. **Public reports** (m_e, j_e) : j_e of m_e reports say closed; each is correct with probability $r \in (\frac{1}{2}, 1)$.

The city publishes the posterior map $\mu_e(s) = \mathbb{P}(\theta_e = 1 \mid s)$ to everyone, so the map itself is public.

Definition 1 (Bayesian Wardrop equilibrium). Given s , flow f^s is an equilibrium if every used path for pair k minimizes $\sum_{e \in P} \tilde{c}_e(x_e^s; s)$ over $\mathcal{P}_k$, where $\tilde{c}_e(x; s) = \mathbb{E}[c_e(x; \theta) \mid s] = a_e x + b_e + M\mu_e(s)$.

3 Results I: the closure map and selfish routing

Lemma 1 (Log-odds fusion). *If $O_e = 1$ then $\mu_e = 1$. Otherwise*

$$\text{logit } \mu_e = \text{logit } \pi_e + \log(1 - \alpha) + k_e \log \frac{p}{q} + (n_e - k_e) \log \frac{1-p}{1-q} + (2j_e - m_e) \log \frac{r}{1-r}. \quad (2)$$

Proof. Independence across edges makes μ_e depend only on edge e 's signals. An alert has zero likelihood when $\theta_e = 0$, so it forces $\mu_e = 1$. Otherwise Bayes' rule with conditional independence gives the posterior odds as prior odds times the three likelihood ratios. These are $(1 - \alpha)/1$ for no alert; $p^k(1 - p)^{n-k}/q^k(1 - q)^{n-k}$ for posts; and $r^j(1 - r)^{m-j}/(1 - r)^j r^{m-j} = (r/(1 - r))^{2j-m}$ for reports (binomial coefficients cancel). Take logs. $\square$

Each channel casts a weighted vote in log-odds space, and official silence is mild evidence that a street is open.

Theorem 1 (Existence and uniqueness [6]). *For every s an equilibrium exists, and its edge loads x^s are the unique minimizer over $\mathcal{F}$ of the Beckmann potential*

$$\Phi_s(x) = \sum_e \int_0^{x_e} \tilde{c}_e(t; s) dt = \sum_e \left(\frac{a_e}{2} x_e^2 + (b_e + M\mu_e) x_e \right).$$

The same holds for any continuous, strictly increasing edge costs, including the density-based costs of Section 6.

Proof. Φ_s is continuous on the compact set $\mathcal{F}$, so a minimizer exists. Each integral is strictly convex in x_e because its derivative $\tilde{c}_e$ is strictly increasing; since x is linear in f , the minimizing load vector is unique. The constraints are linear, so KKT conditions are necessary and sufficient: with multipliers λ_k and $\nu_P \geq 0$, $\sum_{e \in P} \tilde{c}_e(x_e) = \lambda_k + \nu_P$ and $\nu_P f_P = 0$. Thus used paths cost exactly λ_k and no path costs less, which is the equilibrium condition. Conversely, an equilibrium satisfies these KKT conditions with λ_k the common used-path cost. $\square$

Lemma 2 (Bayes consistency). *For any flow rule $f(s)$ depending only on public signals,*

$$\mathbb{E}[C(f(s); \theta)] = \mathbb{E}\left[\sum_e x_e(s) \tilde{c}_e(x_e(s); s)\right].$$

Proof. Condition on s : $x_e(s)$ is s -measurable and $\mathbb{E}[\theta_e | s] = \mu_e(s)$. Apply the tower property. $\square$

Theorem 2 (Price of anarchy on a shared map, cf. [7]). *For any planner flow $f^*(s)$ using the same signals, $\mathbb{E}[C(f^*; \theta)] \leq \frac{4}{3} \mathbb{E}[C(f^*(s); \theta)]$.*

Proof. Fix s ; write $\tilde{c}_e(x) = a_e x + \beta_e$ with $\beta_e \geq 0$ and $\tilde{C}(x) = \sum_e x_e \tilde{c}_e(x_e)$. Let x be equilibrium loads and y feasible. Optimality of x for convex Φ_s gives the variational inequality $\sum_e \tilde{c}_e(x_e) x_e \leq \sum_e \tilde{c}_e(x_e) y_e$. Since $(y_e - x_e/2)^2 \geq 0$,

$$\tilde{c}_e(x_e) y_e = a_e x_e y_e + \beta_e y_e \leq \tilde{c}_e(y_e) y_e + \frac{a_e}{4} x_e^2 \leq \tilde{c}_e(y_e) y_e + \frac{1}{4} x_e \tilde{c}_e(x_e).$$

Summing, $\tilde{C}(x) \leq \tilde{C}(y) + \frac{1}{4} \tilde{C}(x)$, so $\tilde{C}(x) \leq \frac{4}{3} \tilde{C}(y)$. Take $y = x^*(s)$, take expectations, and apply Lemma 2. $\square$

Theorem 3 (Cost of an imperfect map). *Let the city publish the MAP map $\hat{\theta}_e = \mathbf{1}\{\mu_e > \frac{1}{2}\}$ and route to the equilibrium that treats $\hat{\theta}$ as true. Suppose each edge gets at least m (odd) public reports, $\gamma = r - \frac{1}{2}$, and $\bar{C} = D \max_{k, P \in \mathcal{P}_k} \sum_{e \in P} (a_e D + b_e + M)$. Then*

$$\mathbb{P}(\hat{\theta}_e \neq \theta_e) \leq (1 - \alpha \pi_e) e^{-2m\gamma^2}, \quad \mathbb{E}[C(f^{\hat{\theta}}; \theta)] \leq \mathbb{E}[C^{\text{FI}}(\theta)] + \bar{C} e^{-2m\gamma^2} \sum_e (1 - \alpha \pi_e),$$

where C^{FI} is the full-information equilibrium cost.

Proof. Per-edge MAP minimizes per-edge error, so bound the error of the rule “closed if alert, else majority of reports.” If $\theta_e = 0$ it errs only if at least half of m reports are wrong, which by Hoeffding has probability at most $e^{-2m\gamma^2}$. If $\theta_e = 1$ it errs only if no alert $(1 - \alpha)$ and the majority is wrong, at most $(1 - \alpha) e^{-2m\gamma^2}$. Averaging over the prior gives the first bound. On $A = \{\hat{\theta} = \theta\}$ the perceived game equals the true game, so by uniqueness costs coincide; on A^c costs are at most $\bar{C}$. Hence $\mathbb{E}[C] \leq \mathbb{E}[C^{\text{FI}}] + \bar{C} \mathbb{P}(A^c)$, and a union bound finishes. $\square$

Corollary 1. *Any map error has probability at most δ once $m \geq \frac{1}{2\gamma^2} \log \frac{|E|}{\delta}$; e.g. $|E| = 500$, $r = 0.8$, $\delta = 0.05$ needs $m \geq 52$ reports per segment, fewer with alerts and posts.*

4 Results II: coordinated egress needs common knowledge

Selfish routing is a congestion game, but the city’s *plan* is a coordination game. Suppose the MTA and NYPD can open a marshalled corridor to a station and add trains there, which pays off only if fans actually go there, while fans gain only if the corridor is actually open. This is the stag hunt of [1] (their Table 1), relabeled in Table 1.

	Agency: open corridor (T)	Agency: default (A)
Fans: use corridor (T)	h, h	$0, 1$
Fans: default routes (A)	$1, 0$	$1, 1$

Table 1: Egress stag hunt, $h > 1$. With the payoffs of [1], $h = 1.10$. Both (T, T) and (A, A) are equilibria.

Information structure. Let Ω be a finite state space with full-support prior P , and $F \subseteq \Omega$ the event that the corridor plan is feasible (if F fails, T pays 0). Player $i \in \{1, 2\}$ has partition Π_i . For $p \in (0, 1]$, i p -believes E at ω if $P(E \mid \Pi_i(\omega)) \geq p$; write $B_i^p(E)$ for that set. Event E is *evident p -belief* if $E \subseteq B_1^p(E) \cap B_2^p(E)$: whenever it happens, everyone p -believes it. An event is *common p -belief* at ω iff ω lies in some evident p -belief event [2]; for $p = 1$ this is common knowledge in the sense of Lewis and Aumann.

Let $p^* = 1/h$. Because T pays h only if the other plays T and F holds, T is a best response at ω iff

$$h \cdot P(\{\text{other plays T}\} \cap F \mid \Pi_i(\omega)) \geq 1. \quad (3)$$

Theorem 4 (Coordination $\iff$ common p^* -belief). (a) Necessity. *In any Bayesian equilibrium, let $T_i \subseteq \Omega$ be where i plays T. Then $E = T_1 \cap T_2 \cap F$ is evident p^* -belief, so successful coordination only ever happens where F is common p^* -belief.*

(b) Sufficiency. *If $E \subseteq F$ is evident p -belief with $p \geq p^*$, there is a Bayesian equilibrium in which both players play T at every state of E .*

Proof. (a) T_i is a union of Π_i -cells. Fix $\omega \in T_i$. By (3), $P(T_j \cap F \mid \Pi_i(\omega)) \geq p^*$. Since $\Pi_i(\omega) \subseteq T_i$, $P(T_i \cap T_j \cap F \mid \Pi_i(\omega)) = P(T_j \cap F \mid \Pi_i(\omega)) \geq p^*$, so $T_i \subseteq B_i^{p^*}(E)$. Since $E \subseteq T_1 \cap T_2$, $E \subseteq B_1^{p^*}(E) \cap B_2^{p^*}(E)$.

(b) The game is supermodular: i 's gain from T is increasing in the set where j plays T. Let σ^0 have i play T exactly on $S_i^0 = B_i^p(E)$, which is Π_i -measurable. Let BR choose T when indifferent. For $\omega \in S_i^0$, $P(S_j^0 \cap F \mid \Pi_i(\omega)) \geq P(E \mid \Pi_i(\omega)) \geq p \geq p^*$, using $E \subseteq B_j^p(E) = S_j^0$ and $E \subseteq F$. So T is a best response on S_i^0 , i.e. $S_i^0 \subseteq S_i^1 := \text{BR}_i(\sigma^0)$. By supermodularity, $S^t \subseteq S^{t+1}$ implies $S^{t+1} \subseteq S^{t+2}$, so the T-sets increase monotonically. As Ω is finite they stabilize at a profile that is its own best response, an equilibrium with T played on $S_i^\infty \supseteq B_i^p(E) \supseteq E$. $\square$

With $h = 1.10$, $p^* \approx 0.91$: the plan must be near-common knowledge.

Corollary 2 (Which channels can coordinate a plan). *Suppose the plan is announced through channel c , a player who receives it believes the other also received it with probability ρ_c , and a player who did not receive it believes F with probability $\pi_F < p^*$. Then “play T iff I received the announcement” is an equilibrium iff $\rho_c \geq p^*$.*

- *Public broadcast (Wireless Emergency Alert to every phone, arena PA, jumbotron): everyone receives it and sees everyone receive it, so $\rho = 1$; the announcement event is evident 1-belief, i.e. common knowledge, and coordination succeeds.*
- *Social media post: at best secondary knowledge; reach ρ_{social} is typically far below 0.91, so coordination fails.*
- *A private report or direct message: private knowledge, $\rho \approx 0$.*

Proof. A non-recipient’s payoff from T is at most $h\pi_F < 1$, so A is strictly optimal. A recipient’s payoff from T is $h\rho_c$ (the other plays T exactly when they received it, and F holds when announced), which is ≥ 1 iff $\rho_c \geq p^*$. $\square$

This reproduces the ordering in Fig. 2 of [1]: coordination rises from private to shared knowledge and peaks with a broadcast. In Rubinstein’s email game [3], any *finite* depth of “I know you know...” still fails to sustain coordination, which is why the official plan must be broadcast, not relayed.

5 Results III: who reports the closure?

The map in Theorem 3 needs reports, and reporting is the volunteer’s dilemma [4] that [1] use for the bystander effect. Each of N fans who see a closure may report at cost κ ; if at least one reports, all gain $B > \kappa$.

Proposition 1 (Bystander effect under common knowledge). *If the closure and the set of witnesses are common knowledge, the unique symmetric equilibrium has report probability $p_N = 1 - (\kappa/B)^{1/(N-1)}$, and $\mathbb{P}_N(\text{no report}) = (\kappa/B)^{N/(N-1)}$ is strictly increasing in N , tending to κ/B .*

Proof. Pure symmetric profiles are not equilibria (if nobody reports, deviating gains $B - \kappa > 0$; if everyone reports, deviating saves κ). Indifference $B - \kappa = B(1 - (1 - p)^{N-1})$ gives $(1 - p)^{N-1} = \kappa/B$, so p_N is unique. Then $\log \mathbb{P}_N = \frac{N}{N-1} \log \frac{\kappa}{B}$, with $\log(\kappa/B) < 0$ and $\frac{N}{N-1} \downarrow 1$, so it increases to $\log(\kappa/B)$. $\square$

Proposition 2 (Knowledge-level zigzag). *Two fans witness a closure. A fan at level 1 (private knowledge) believes the other is unaware; a fan at level $n \geq 2$ believes the other is at level $n - 1$, as in the messenger design of [1]. An unaware fan never reports. Then a fan at finite level n reports iff n is odd.*

Proof. Induction on n . Level 1: the other is unaware and will not report, so reporting gives $B - \kappa > 0$ versus 0; report. Level $n \geq 2$: the other is at level $n - 1$; by induction they report iff $n - 1$ is odd. If so, not reporting yields $B > B - \kappa$; do not report. If $n - 1$ is even, the other does not report, so reporting yields $B - \kappa > 0$; report. Hence report iff n is odd. $\square$

This is the private/secondary/tertiary zigzag found in Fig. 3 of [1], while the bystander effect of Proposition 1 appears only under common knowledge. Two design rules follow. *Break the symmetry*: if the app privately pings one witness (“you’re the closest; please confirm”), that fan is at level 1 and reports with certainty. *Pay for reports*: a reward $\rho \geq \kappa$ makes reporting weakly dominant for all.

6 Results IV: density-aware routing prototype

The accompanying script `knicks_router.py` implements the model on a Midtown grid (5th–10th Ave, W 26th–W 45th St) with exits around the Garden and five transit destinations.

Density map. A Gaussian kernel density estimate of (simulated) phone pings around celebration hot spots gives a crowd density field $\rho_0(x, y)$ in persons/m². Routed fans add density $\Delta\rho_e(f_e) = f_e \tau_e / (T L_e w)$, where τ_e is free-walk time, T the egress window, L_e length, and w walkable width.

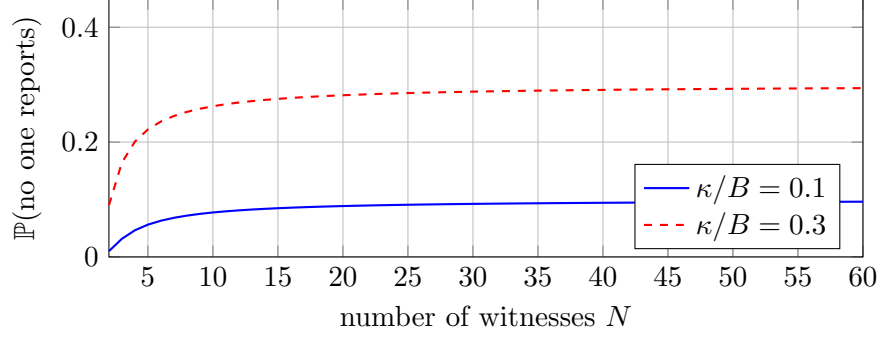


Figure 1: Proposition 1: under common knowledge, larger crowds leave closures unreported more often.

Edge cost. Walking speed follows the Greenshields fundamental diagram $v(\rho) = v_f(1 - \rho/\rho_{\text{jam}})$ (floored at $v_{\min}$), with $v_f = 1.34 \text{ m/s}$ and $\rho_{\text{jam}} = 5.4 \text{ persons/m}^2$, so

$$c_e(f_e) = \frac{L_e}{\max\{v_{\min}, v_f(1 - (\rho_{0,e} + \Delta\rho_e(f_e))/\rho_{\text{jam}})\}} + M\mu_e.$$

This is continuous and increasing in f_e , so Theorem 1 guarantees a unique equilibrium. (The 4/3 bound of Theorem 2 is specific to affine costs.)

Closure map and assignment. Signals are simulated per street and fused by Lemma 1. Equilibrium is computed by the method of successive averages, which converges for separable increasing costs [8]; the final relative gap is 0.003%.

Routing strategy	trip (min)	into barricades	blocks at $> 2 \text{ p/m}^2$
Naive shortest path	24.8	13,200	40,000
Closure-aware (fused signals)	15.7	0	25,980
Closure + density-aware equilibrium	15.4	0	19,975

Table 2: Synthetic evaluation (20,000 fans, seed 7), scored under the true closures. The fused map detected all 14 directed closed segments with 2 false alarms. Density awareness barely changes trip time but cuts exposure to crush-risk densities by 23% versus closure-only and 50% versus naive routing.

7 Discussion

Could official alerts, social media updates, and public reports help map street closures and plan alternate routes home? Yes, with a division of labor dictated by levels of knowledge:

1. **Mapping** uses all three channels, fused by Lemma 1; errors vanish exponentially in report volume (Theorem 3).
2. **Routing** works on a *shared* map, with a unique equilibrium within 4/3 of optimal (Theorems 1, 2), and density-aware costs reduce crowd crush risk.

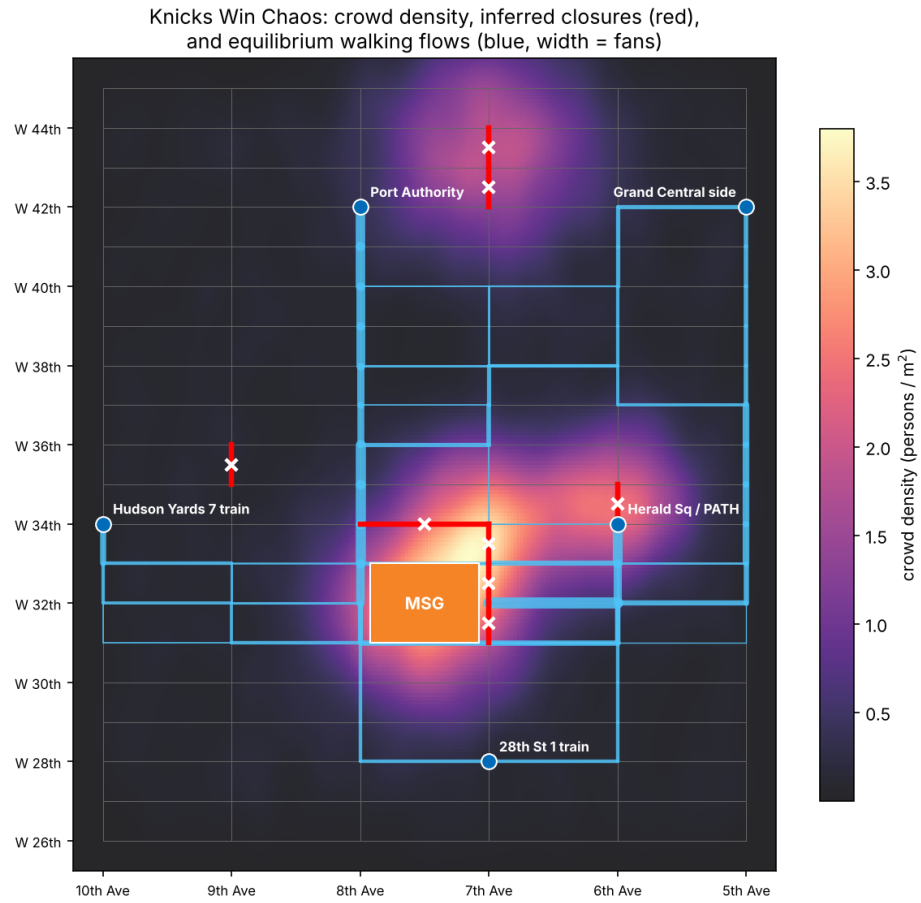


Figure 2: Output of `knicks_router.py`: crowd density (heat map), inferred closures (red), and equilibrium walking flows (blue, width proportional to fans).

3. **Coordinated plans** (corridors, extra trains) must be *broadcast* to create common knowledge (Theorem 4, Corollary 2); social media alone cannot do it.
4. **Reporting** must be designed around the volunteer’s dilemma: ping individual witnesses or reward reports (Propositions 1, 2).

Limitations. Signals are assumed conditionally independent (retweets violate this); closures are exogenous, though crowds create them; the knowledge-level model of Proposition 2 is the stylized hierarchy of [1]; and the simulation uses synthetic data and approximate block lengths.

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